

Rings and Modules

1. Find an example of each of the following.

- (a) (**Aug 2018 1(a)**) A short exact sequence of modules that is not split exact.
- (b) A projective module that is not free.
- (c) A flat module that is not projective.
- (d) (**Aug 2018 1(b)**) and (**Aug 2020 1(d)**) A flat $\mathbb{Z}$ -module that is not free.
- (e) Nonzero modules M, N over a ring R such that $M \otimes_R N = 0$.
- (f) A ring R and an R -algebra A that is finitely generated as an algebra over R but is not finitely generated as an R -module.

1. Solution

- (a) Let $R = \mathbb{Z}$ and consider the SES of $\mathbb{Z}$ -modules $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z} \rightarrow 0$ where the left map is multiplication by 3 and the right map is the natural quotient map. This is not split exact since $\mathbb{Z} \not\cong \mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$.
- (b) Consider $M = \mathbb{Z}/2\mathbb{Z}$ as a module over $R = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$. It is projective since it is a direct summand of R , but it is not a free R -module.
- (c) Consider $\mathbb{Q}$ as a $\mathbb{Z}$ -module. It is flat because localization is flat, but it is not projective since over a PID being projective is equivalent to being free and $\mathbb{Q}$ is not a free $\mathbb{Z}$ -module.
- (d) Since $\mathbb{Z}$ is a PID, a module is free iff it is projective, so this is really the same as the previous question.
- (e) Let $R = \mathbb{Z}$, $M = \mathbb{Z}/2\mathbb{Z}$, and $N = \mathbb{Z}/3\mathbb{Z}$. Then $M \otimes_R N = \mathbb{Z}/\gcd(2, 3)\mathbb{Z} = 0$.
- (f) Let $R = \mathbb{C}$ and $A = \mathbb{C}[x]$. Then A is generated as an R -algebra by 1 and x , but as an R -module it is generated by $1, x, x^2, \dots$.

2. (**August 2017 Problem 1**) Let $M = \mathbb{C}[x]/(x^2 + x)$ and $N = \mathbb{C}[x]/(x - 1)$. Compute the dimension of the following tensor products as vector spaces over $\mathbb{C}$:

- (a) $M \otimes_{\mathbb{C}[x]} N$
- (b) $M \otimes_{\mathbb{C}} N$
- (c) $M \otimes_{\mathbb{C}[x^2]} N$.

2. Solution

- (a) Recall that $R/I \otimes_R R/J \cong R/(I + J)$, so $M \otimes_{\mathbb{C}[x]} N \cong \mathbb{C}[x]/(x - 1, x^2 + x) = 0$. Thus, it is 0-dimensional over $\mathbb{C}$.
- (b) We have $M \otimes_{\mathbb{C}} N \cong M \otimes_{\mathbb{C}} \mathbb{C} \cong M$, which is 2-dimensional over $\mathbb{C}$ with basis $1, x$.

- (c) The only possible basis elements are $1 \otimes 1$ and $x \otimes 1$. Since we can multiply across the tensor by x^2 , we have $x \otimes 1 = -x^2 \otimes 1 = -(1 \otimes x^2) = -(1 \otimes 1)$. So this is 1-dimensional over $\mathbb{C}$ with basis $1 \otimes 1$.

3. (January 2019 Problem 1)

- (a) Give an example of a simple ring which is not a commutative ring.
- (b) Give an example a ring R , R -modules M_1, M_2, M_3 , and N and a short exact sequence $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ such that the corresponding sequence $0 \leftarrow \text{Hom}(M_1, N) \leftarrow \text{Hom}(M_2, N) \leftarrow \text{Hom}(M_3, N) \leftarrow 0$ is not exact.

3. Solution

- (a) Let R be the ring of 2×2 matrices over $\mathbb{F}_2$. Then R is not commutative, and R is simple since the only nonzero two-sided ideal is R .
- (b) Let $R = \mathbb{Z}$ and consider $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$. Then applying $\text{Hom}(-, \mathbb{Z}/2\mathbb{Z})$ gives

$$0 \rightarrow \text{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \rightarrow \text{Hom}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z} \rightarrow \text{Hom}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z} \rightarrow 0$$

which is not exact on the right since the map is multiplication by $2 = 0$, so it is not surjective.

4. (August 2020 Problem 4) Let R be a commutative ring.

- (a) Give an example of a free R -module M and an injective homomorphism $M \rightarrow M$ which is not an isomorphism. (For this part, R can be whatever commutative ring you like.)
- (b) If M_1 and M_2 are free of finite ranks r_1 and r_2 , show that a surjective map $M_1 \rightarrow M_2$ exists if and only if $r_1 \geq r_2$.
- (c) Assume furthermore that R is an integral domain. Show that an injective map $M_1 \rightarrow M_2$ exists if and only if $r_1 \leq r_2$.

4. Solution

- (a) Recall that if M is finitely generated as an R -module, then any surjective endomorphism is injective. This problem shows that the converse is not true. For example, consider $\mathbb{Z}$ as a module over itself. Then the map $\mathbb{Z} \rightarrow \mathbb{Z}$ given by multiplication by 2 is injective but not surjective.
- (b) We have $M_1 \cong R^{r_1}$ and $M_2 \cong R^{r_2}$. Consider a map $\varphi : R^{r_1} \rightarrow R^{r_2}$, and let $\mathfrak{m}$ be a maximal ideal in R so that $R/\mathfrak{m} \cong k$ where k is a field. It is a fact that φ is surjective iff $\varphi_{\mathfrak{m}}$ is surjective for every maximal ideal $\mathfrak{m}$, so we may assume that R is a local ring with unique maximal ideal $\mathfrak{m}$. Then consider the commutative diagram

$$\begin{array}{ccc} R^{r_1} & \xrightarrow{\varphi} & R^{r_2} \\ \downarrow & & \downarrow \\ k^{r_1} & \xrightarrow{\varphi} & k^{r_2} \end{array}$$

where the vertical maps are the natural quotient maps and $\bar{\varphi}$ is induced by φ . By Nakayama's Lemma, we know that $g_1, \dots, g_{r_2}$ are generators for M_2 iff $\bar{g}_1, \dots, \bar{g}_{r_2}$ are generators for k^{r_2} . Then φ is surjective iff there exists $f_i \in R^{r_1}$ such that $\varphi(f_i) = g_i$ for each $i = 1, \dots, r_2$. And this occurs iff there exists $\bar{f}_i \in k^{r_1}$ such that $\bar{\varphi}(\bar{f}_i) = \bar{g}_i$ for each $i = 1, \dots, r_2$. Since $\bar{\varphi}$ is a map of vector spaces, this happens iff $r_1 \geq r_2$.

- (c) Since R is a domain, (0) is a prime ideal, so let $k = \text{Frac}(R) = R_{(0)}$. Any map $\varphi : R^{r_1} \rightarrow R^{r_2}$ induces a map $\varphi_{(0)} : k^{r_1} \rightarrow k^{r_2}$, which is a map of vector spaces. Since localization is exact, φ is injective iff $\varphi_{(0)}$ is injective, which happens iff $r_1 \leq r_2$.

5. (January 2022 Problem 3) Let R be a commutative ring.

- (a) Prove that a module M over R is projective if and only if it is a direct summand of a free module.
- (b) Let M be a projective module over R and $S \subset R$ a multiplicative set. Prove that $S^{-1}M$ is a projective $R[S^{-1}]$ -module.

5. Solution

- (a) For the forward direction, assume M is projective, and let $\pi : F \rightarrow M$ be a surjective map where F is a free module. Such a map exists since every module is a homomorphic image of a free module. Then by definition of M being projective, there exists a map $\varphi : M \rightarrow F$ such that $\pi \circ \varphi = id_M$. Therefore, the short exact sequence $0 \rightarrow \ker \pi \rightarrow F \xrightarrow{\pi} M \rightarrow 0$ splits, and thus $F \cong M \oplus \ker \pi$.

For the reverse direction, assume $F \cong M \oplus N$ for some free module F . Then there exist natural maps $\alpha : M \rightarrow F$ and $\beta : F \rightarrow M$ given by inclusion and projection such that $\beta \circ \alpha = id_M$. Suppose we have a surjective map $\pi : L \rightarrow K$ and a map $\psi : M \rightarrow K$. We need to show that there exists a map $\varphi : M \rightarrow L$ such that the following commutes:

$$\begin{array}{ccc} & M & \\ \swarrow \varphi & \downarrow \psi & \\ L & \xrightarrow{\pi} & K \end{array}$$

Since F is free, it is projective (this is not hard to prove using that F has a basis and so any map out of F is defined by what it does to the basis elements). Therefore, considering the map $\psi \circ \beta : F \rightarrow M$, we know that there exists a map $\gamma : F \rightarrow L$ such that $\pi \circ \gamma = \psi \circ \beta$. Then defining $\varphi = \gamma \circ \alpha : M \rightarrow L$, we see that $\pi \circ \varphi = \pi \circ \gamma \circ \alpha = \psi \circ \beta \circ \alpha = \psi$. Thus, M is projective.

- (b) Since M is projective, by part (a), we have that $M \oplus N \cong R^d$ for some R -module N and some $d \geq 1$. Therefore, we have a split SES $0 \rightarrow N \rightarrow R^d \rightarrow M \rightarrow 0$. Since localization is exact, we get that $0 \rightarrow S^{-1}N \rightarrow S^{-1}(R^d) \rightarrow S^{-1}M \rightarrow 0$ is also a split SES. Thus, $S^{-1}M \oplus S^{-1}N \cong S^{-1}(R^d) \cong R[S^{-1}]^d$, so $S^{-1}M$ is a projective $R[S^{-1}]$ -module by part (a).

6. Show that if R is a PID (principal ideal domain) and M is an R -module, then M is flat if and only if M is torsion free.

6. Solution We will use the fact that M is flat iff for any ideal $I \subseteq R$, the natural map $I \otimes_R M \rightarrow R \otimes_R M \cong M$ is injective. Since R is a PID, we know that every ideal $I = (f)$ for some $f \in R$. We want to show that the map $\varphi : I \otimes_R M \rightarrow M$ given by $i \otimes m \mapsto im$ is injective for any I . If $f = 0$, then this is clear, so assume that $f \neq 0$. Observe that every element of $I \otimes_R M$ can be written in the form $f \otimes m$ for some $m \in M$. This is because $r_1 f \otimes m_1 + r_2 f \otimes m_2 = f \otimes (r_1 m_1 + r_2 m_2)$. Let $f \otimes m \in \ker \varphi$ so that $\varphi(f \otimes m) = fm = 0$ in M . Then we see that φ is injective for all possible $f \in R$ iff M is torsion free.

7. (January 2021 Problem 3)

- (a) Show that a projective module over an integral domain is torsion free.
- (b) Let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence of finitely generated modules over a commutative, Noetherian ring. Prove the following OR give a counterexample:
 - (i) If M_1 and M_2 are projective, then so is M_3 .
 - (ii) If M_2 and M_3 are projective, then so is M_1 .

7. Solution

- (a) Let M be a projective R -module, where R is an integral domain. Then M is a direct summand of a free R -module, i.e. $M \oplus N = R^d$ for some R -module N and $d \in \mathbb{N}$. Let $m \in M$ be a nonzero element, and suppose that $am = 0$ for some $a \in R$. We will show that $a = 0$. We can write $m = (r_1, \dots, r_d)$ with $r_i \in R$. Then $am = (ar_1, \dots, ar_d) = 0$ implies that $ar_i = 0$ in R for all i . Since R is a domain, this means that either $a = 0$ or $r_i = 0$ for all i . Since $m \neq 0$, we must have that $a = 0$. Thus, M is torsion free.
- (b) (i) This is not true. For example, consider the SES of $\mathbb{Z}$ -modules $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$, where the left map is multiplication by 2 and the right map is the natural quotient map. Then $\mathbb{Z}$ is projective, but $\mathbb{Z}/2\mathbb{Z}$ is not a projective $\mathbb{Z}$ -module. One way to see this directly is that if $\pi : \mathbb{Z}/4\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is the map given by modding out by 2 and $\mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is the identity map, then there is no map that makes the following

diagram commute:

$$\begin{array}{ccc}
 & \mathbb{Z}/2\mathbb{Z} & \\
 & \swarrow \text{dashed} & \downarrow \\
 \mathbb{Z}/4\mathbb{Z} & \longrightarrow & \mathbb{Z}/2\mathbb{Z}
 \end{array}$$

This because the only possible maps are the

zero map and multiplication by 2, and neither of these gives the identity map when composed with π . Another way to see that $\mathbb{Z}/2\mathbb{Z}$ is not a projective $\mathbb{Z}$ -module is to use that every projective module is flat and show that it is not flat.

- (ii) This is true. If M_3 is projective, then the SES is split exact, so $M_2 \cong M_1 \oplus M_3$. Recall that being projective is equivalent to being a direct summand of a free module. If M_2 is projective, then M_1 is as well since it is a direct summand of M_2 which is a direct summand of a free module.

8. (August 2021 Problem 4) Decide if the following statements are true. If so, provide a proof, if not, provide a counterexample. Let M, N be finitely generated modules over a commutative ring A . Let $f: M \rightarrow N$ be a homomorphism of A -modules, and let $g: B \rightarrow A$ be a homomorphism of commutative rings.

- (a) “If f is injective, so is the induced morphism $M \otimes_A M \rightarrow N \otimes_A N$.”
- (b) “If f is surjective, so is the induced morphism $M \otimes_A M \rightarrow N \otimes_A N$.”
- (c) “If g is injective, so is the induced morphism $M \otimes_B M \rightarrow M \otimes_A M$.”

8. Solution

- (a) We should suspect this is false because tensor products usually don’t play well with injections. So we’re going to need either M or N to be non-flat, but how do we figure out which one we need?

The morphism $M \otimes_A M \rightarrow N \otimes_A N$ factors as $M \otimes_A M \rightarrow M \otimes_A N \rightarrow N \otimes_A N$. If the composition were injective, then in particular the first map $\text{id} \otimes f$ would also be injective. However, we know $\text{id} \otimes f$ need not be injective if M is not flat, so that gives us an avenue to look for a counterexample.

Here’s one possible counterexample: let $A = \mathbb{Z}$, $M = \mathbb{Z}/2\mathbb{Z}$, $N = \mathbb{Z}/4\mathbb{Z}$, and $f: M \rightarrow N$ given by $f(1) = 2$, which is injective. Then we have that $M \otimes_A M \cong \mathbb{Z}/2\mathbb{Z}$ where the nonzero element is $1 \otimes 1$, $N \otimes_A N \cong \mathbb{Z}/4\mathbb{Z}$, but the induced map sends $1 \otimes 1$ to $2 \otimes 2 = 4 \otimes 1 = 0 \times 1$, so the induced map is the zero map.

- (b) This one is true. Looking at the factorization from part (a), both $\text{id} \otimes f$ and $f \otimes \text{id}$ are surjective, since tensor products preserve surjections. But then the composition of two surjections is a surjection.
- (c) This one is mega false. For example, let $A = \mathbb{C}$, $B = \mathbb{R}$, and $M = \mathbb{C}$. Then $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \rightarrow \mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$ is not injective, because the one on the left has real dimension 4 and the one on the right has real dimension 2. In fact, I think quite to the contrary that $M \otimes_B M \rightarrow M \otimes_A M$ is surjective when $B \rightarrow A$ is injective, and visa versa.

9. (August 2017 Problem 5) (This problem has been slightly modified from its original form.) Let $B = \mathbb{Q}[x, y]/(xy)$.

- (a) Consider the map $\mathbb{Q}[t] \rightarrow B$ sending t to x . This map makes B into a $\mathbb{Q}[t]$ -module. Show that B is *not* a finite $\mathbb{Q}[t]$ -module under this map.
- (b) Consider the map $\mathbb{Q}[t] \rightarrow B$ sending t to $x + y$. Show that this map makes B into a finite $\mathbb{Q}[t]$ -module.
- (c) Consider B as a $\mathbb{Q}[t]$ -module under the map from part (a). Is B a flat $\mathbb{Q}[t]$ -module? What about when considered as a $\mathbb{Q}[t]$ -module under the map from part (b).

9. Solution

- (a) Suppose B were finitely generated, with generating set $\{f_1, \dots, f_n\}$. Then each of the f_i has finite degree in the variable y , so there is a maximum degree, say y^k . But then the y -degree of a $\mathbb{Q}[t]$ -linear combination $a_1 f_1 + \dots + a_n f_n$ is at most k , so there is no way to write y^{k+1} (or any higher power) in this way. Thus, B cannot have a finite generating set.
- (b) We will show that under this map, B has generating set $\{1, x\}$. All we need to show is that any power x^k or y^k can be written as a $\mathbb{Q}[t]$ -linear combination of these two elements. We have x already in our generating set, and for $k > 1$ we have $x^k = (x+y)x^{k-1} = tx^{k-1}$. On the other hand, we have $y^k = (x+y)^k - x^k = t^k - x^k$.
- (c) The ring $\mathbb{Q}[t]$ is a PID. Recall from problem 6 that a module over a PID is flat iff it is torsion-free. In part (a), B has torsion because $ty = xy = 0$. However, in part (b) B is torsion-free, hence flat.